Sales & Operations Intelligence Dashboard

Product Requirements Document (PRD)

Intern assignment • Timebox: 2 days • Prototype only

1. Assignment Overview

Build a small, hosted prototype of a dashboard that combines sales and operational data from multiple sources and presents useful reports for sales and management users. The intended long-term tool would connect to CRM, 3rd Party operations system, Google Sheets, and many more. The candidate will not receive access to these live systems or company data.

For this two-day assignment, use synthetic/sample datasets that resemble realistic business records. The goal is to assess AI and data skills, analytical thinking, dashboard usability, technical choices and the candidate's plan for a future secure integration.

2. Problem Statement

Information needed for sales and management decisions is held across separate systems. Manual consolidation can be slow, inconsistent and difficult to repeat. This prototype should demonstrate how data from different sources can be combined, checked, analysed and presented as useful dashboards - without relying on live company data.

3. Objectives

- Build a working dashboard using synthetic data representing CRM, operations and spreadsheet sources.
- Demonstrate a repeatable way to load and combine multiple datasets.
- Present a practical selection of KPIs, charts, trends and filters.
- Explain the technology stack and the path to a production implementation.

4. Scope and Constraints

4.1 In scope

- Synthetic datasets in Google Sheets, JSON source and a lightweight database.
- At least three separately identifiable datasets representing CRM, OPS system and Google Sheets.
- Sales and management/operations dashboard views.
- Basic data mapping, joins, normalization and data-quality handling.
- At least two working filters and one AI-assisted or rule-based insight.
- Hosted demo, source repository, video and concise documentation.

4.2 Out of scope

- Connecting to live company accounts or using real client, prospect, employee, financial or operational data.

- Implementing every possible report or field.
- Building a production-grade data warehouse or enterprise permissions system.
- Claiming production security, complete data accuracy or validated AI forecasting.

5. Data Sources and Sample Data

Create realistic but entirely synthetic data for all three source categories. Clearly label each dataset and document its fields, meaning, units, date formats, missing values and relationships.

Represented source	Possible sample fields	Purpose
CRM	Lead/contact ID, created date, source, owner, company, region, pipeline stage, deal value, status, last activity	Lead funnel, pipeline and sales performance
Operations	Project/order ID, client ID, status, service type, created/due/completed dates, amount, assigned team, SLA status	Workload, delivery and operational performance
Google Sheets	Targets/forecast, account segment, campaign mapping, manual category or reference values	Targets, mappings and management context

Do not assume lead, contact, client and project IDs are interchangeable across systems. Define a reasonable mapping/join strategy and document any assumptions or unmatched records.

6. Functional Requirements

6.1 Data loading and integration simulation

- Load data from three separate sample-source datasets.
- Provide a simple repeatable way to load or refresh the data, such as a refresh button or documented reload process.
- Normalize important fields such as dates, statuses, amounts and identifiers.
- Handle missing, duplicate or unmatched records sensibly; do not silently hide major data-quality issues.
- Show the data load/refresh time or clearly label the data as static sample data.

6.2 Sales dashboard

Implement a useful subset supported by the sample data:

- Leads created over time.
- Lead distribution by source, pipeline stage, owner or region.
- Open pipeline/deal value and status breakdown, if the required fields exist.
- Stage conversion only when stages and denominators are clearly defined.
- Stale-lead or follow-up indicators if activity data is available.

6.3 Management / operations dashboard

- Project/order volume by status and over time.
- Open, overdue and completed work based on documented date/ timelines/ SLA rules.
- Workload by service type, client or assigned team.
- A combined sales-to-operations view where accounts can be matched defensibly.
- Targets versus actuals or forecast versus actuals if sample target data is included.

6.4 Filters and exploration

- Include at least two useful filters (e.g. date range, owner, source, region, client, service or status).
- Ensure relevant charts and KPIs update consistently when filters change.

- Where practical, allow the user to inspect records behind a KPI or chart.
- Use clear labels, date ranges, units and KPI definitions.

6.5 AI-assisted insights

Include at least one working AI-assisted insight. If an AI service is not feasible within the timebox, a clearly identified analytical/rule-based insight is acceptable, but the candidate must explain the method and trade-off.

- Examples: a change in lead volume, pipeline concentration, overdue work, target/actual gap or records requiring review.
- Tie every insight to a visible metric or sample records and explain why it may matter.
- Separate observed facts from possible explanations or suggested actions.
- If an LLM is used, send only synthetic data and do not let it invent missing values, causes or metrics.
- Do not present an unsupported forecast or causal explanation as fact.

7. Example Questions the Dashboard Should Help Answer

- How many leads were created during the selected period, and by which sources or stages?
- Which stages or owners contain the largest amount/number of open opportunities, based on available fields?
- How many operational projects are open, overdue or completed?
- Which clients/accounts can be matched across sales and operations data?
- What data trend or exception should a manager investigate next including comparison with previous years data of some time frame?

8. Data Quality and Trust

- Provide a data dictionary and document KPI calculation rules.
- Document join keys, matching rules and assumptions across sources.
- Avoid double-counting records that appear multiple times or in multiple datasets.
- Report unmatched records and important missing data.
- Use consistent date ranges, time zones and currency units; document or exclude cross-currency totals.
- Do not calculate a KPI if the sample data cannot support it reliably.

9. Security and Privacy

- Use synthetic data only. Do not request company credentials, exports, screenshots or live records.
- Never commit API keys, access tokens, passwords or other secrets to Git.
- Document where data is stored and whether any external AI service receives it; only synthetic data may be sent.
- Ensure the hosted demo does not expose secrets or private data.
- Describe privacy, retention, access control and audit requirements for production.

10. Acceptance Criteria

The evaluator should be able to:

1. Open the hosted app and see sample data representing all three source categories.
2. View a sales dashboard and a management/operations dashboard or clearly separated views.
3. Use at least two filters and see relevant charts/KPIs update.
4. Understand how datasets were mapped and inspect documented assumptions.

5. View at least one AI-assisted or analytical insight linked to the sample data.
6. Find KPI definitions, sample-data limitations and refresh/load status.
7. Open the Git repository and understand how to run the application locally.

A small, accurate and explainable dashboard is preferable to many charts with unclear definitions or incorrect joins.

11. Technology Stack and Architecture

The candidate may choose any reasonable stack. The submission must explain the frontend/dashboard framework, backend or data-processing layer, storage approach, charting library, AI/model approach and hosting platform. Explain why these choices suit a two-day prototype, their trade-offs and how the design could evolve.

Using sample files is acceptable. Do not spend the assignment trying to connect to GHL, Plunet or Google Sheets accounts that have not been provided.

12. Production-Readiness Vision

Provide a concise roadmap explaining what would be required to turn the prototype into a dependable internal tool. Address:

- Integration options for GHL (CRM) and Google Sheets: APIs/exports, authentication, permissions, rate limits and provider constraints.
- Repeatable ingestion/synchronization, incremental updates, retries, error handling and monitoring.
- Canonical data model, identifier mapping, data dictionary, quality checks and reconciliation.
- Database/data warehouse needs and consistent KPI calculation ownership.
- Role-based access for sales users, managers and administrators.
- Privacy, retention/deletion, encryption, secret management, audit logs and backups.
- Testing for joins, KPI calculations, freshness, access control and regressions.
- AI governance: evaluation, traceability to data, uncertainty, human review and safeguards against invented conclusions.
- Deployment, CI/CD, monitoring, performance, maintenance, costs and support ownership.
- Phased roadmap, dependencies, risks and indicative effort.

13. Required Submission Deliverables

- Publicly hosted working prototype URL accessible to the evaluator without company-system access.
- Git repository link with source code, README, setup instructions, architecture summary, data dictionary and limitations.
- Short screen-recorded demo video (approximately 3–5 minutes) showing dashboards, filters, data mapping and insight.
- Technology-stack explanation and rationale.
- Production-readiness roadmap.
- Brief list of assumptions, completed/incomplete features, known defects and AI/API dependencies.
- Synthetic sample data files or a documented way to generate them.

Deadline: submit all deliverables within 2 days of receiving the assignment. This is a scoped prototype, not a full MVP or production-ready application.

14. Evaluation Criteria

Criterion	What will be assessed	Weight
Dashboard and usability	Hosted demo, useful views, clear layout and working filters	20%
Data integration and modelling	Three source representations, joins/mapping, normalization and refresh approach	20%
AI / analytical capability	Insight linked to data, reasoning and uncertainty handling	20%
KPI accuracy and data quality	Definitions, calculations, duplicates/missing/unmatched records	15%
Engineering and technical decisions	Maintainability, error handling, suitable stack and readable code	10%
Security and privacy	Synthetic data, secrets handling and production safeguards	5%
Production-readiness thinking	Practical roadmap, integration realities and risks	5%
Submission quality	Git, README, hosted URL, video and documentation	5%

15. Important Clarifications

- No access to live GHL, 3rd Party Systems, company Google Sheets or company data will be provided.
- Synthetic/sample data is mandatory; generating realistic sample data is part of the assignment.
- Live integration with all three systems is not expected in two days.
- AI should support analysis, not merely decorate the dashboard. Insights must be traceable to data.
- Do not claim production readiness or validated AI performance from a prototype.
- Prioritize correctness, clarity and credible technical decisions over feature quantity.

16. Candidate Submission Checklist

- ☐ Hosted demo URL tested
- ☐ Git repository and README complete
- ☐ Demo video link included
- ☐ Synthetic datasets and data dictionary included
- ☐ Stack and rationale documented
- ☐ KPI definitions and data mapping explained
- ☐ At least one insight demonstrated
- ☐ Production roadmap included
- ☐ Assumptions and limitations documented
- ☐ No company data or secrets exposed